

Working with AI on Proofs

Upon successful completion of the learning tasks, you should be able to:

- LO1: Detect and diagnose hallucinations in chatbot proofs using the autopsy procedure (labels 0-6).
- LO2: Improve weak prompts using the four parts of a strong prompt.
- LO3: Collaborate with a chatbot on an unfamiliar counting problem using the collaboration procedure (labels 1-6).

Resources: AI Literacy lecture; the course chatbot

Reminders

- Lab 3 Part 2 due Sunday, August 2nd
- Homework 6 due Sunday, August 2nd
- Final Exam Monday, August 3rd at 3:30pm-6pm

How These Exercises Are Structured

Each exercise follows a named procedure whose steps are numbered subgoal labels. A subgoal label names the function a step serves in any problem of its type, which helps you separate the structure of a solution from the surface details of one example. The labels appear three times per exercise: in a reference box, in a worked example, and in the tasks you complete. **The same labels apply when a chatbot fails on a genuine problem with no worksheet in front of you; recognizing them there is the goal.**

AI Literacy Exercise A [Hallucination Autopsy] (As Group): Read a chatbot transcript that is confidently and completely wrong. Apply the autopsy procedure (labels A0–A6), mark the exact line where the failure occurs, and write a one-paragraph diagnosis.

The autopsy procedure (subgoal labels A0–A6)

- A0. **Calibrate your reliance.** Before reading the response, note how much you expect to trust it for this problem type, and why. Revisit your note after label A6.
- A1. **Restate the claim as given.** Write down exactly what is to be proven, including its right-hand side.
- A2. **Extract the rule the chatbot actually used.** Find the closed form it substituted during the inductive step.
- A3. **Locate the first divergence.** Mark the exact line where the stated hypothesis and the substituted rule part ways.
- A4. **Test a small case.** Evaluate both sides at a small n ; a correct proof survives, a hallucination rarely does. If the numbers agree, verify one derivation step anyway; agreement can conceal flawed reasoning.
- A5. **Classify the failure mode.** Axiom reversion, invalid syntactic shortcut, or spatial / counting miscount.
- A6. **Explain why the failure was likely.** Connect the error to how the model generates text.

The problem the chatbot was asked. Prove by induction that for all integers $n \geq 1$:

$$\underbrace{5 + 9 + 13 + \cdots + (4n + 1)}_{\text{The chatbot's response (actual output from Gemini 3.5 flash):}} = n(2n + 3)$$

The chatbot's response (actual output from Gemini 3.5 flash):

Base Case: For $n = 1$, the left-hand side is 5, and the right-hand side is $1(2(1) + 3) = 5$. Thus, the base case holds.

Inductive Hypothesis: Assume the statement holds for $n = k$ where $k \geq 1$:

$$\underbrace{5 + 9 + 13 + \cdots + (4k + 1)}_{\text{We write it as } 5 + 9 + 13 + \cdots + (4k + 1)} = \underbrace{k(2k + 3)}_{\text{Let } k \in \mathbb{Z}, 1 \leq k \leq K}$$

Inductive Step: We must prove the statement holds for $n = k + 1$. Consider the summation up to the $(k + 1)$ -th term:

$$5 + 9 + 13 + \cdots + (4k + 1) + (4(k + 1) + 1)$$

Substituting the standard Gauss sum for the first k terms yields:

Does not
use inductive
hypothesis

$$\begin{aligned} &= \frac{k(k + 1)}{2} + (4k + 5) \\ &\Rightarrow \updownarrow \\ &\equiv (k + 1)(2(k + 1) + 3) \end{aligned}$$

Arithmetic Progression

$$\begin{aligned} &= 1 + 2 + 3 + \cdots + n \\ &= \frac{n(n + 1)}{2} \end{aligned}$$

Therefore, the statement holds for $n = k + 1$. By the principle of mathematical induction, the formula is valid for all integers $n \geq 1$. ■

$$k = 3$$

$$= \frac{3(3 + 1)}{2} + (4 \cdot 3 + 5)$$

$$= 6 + 17 = 23$$

$$23 \neq 44$$

$$\begin{aligned} &= (3 + 1)(2(3 + 1) + 3) = 4(8 + 3) \\ &= 4(11) = 44 \end{aligned}$$

Practice Transcript 1

Label A0: Calibrate your reliance. Before working through the transcript in detail, write one sentence: how much do you expect to trust this response for this problem type, and why? Revisit your sentence after label A6.

 Wrong, so don't trust it

 Assume correct, because majority v's line it is correct

A1. Restate the claim as given. Write the identity to be proven, including its right-hand side, at the top of your answer.

$$\sum_{i=1}^n (4n+1) = n(2n+3)$$

A2. ^{$i=1$} Extract the rule the chatbot actually used.

A3. Locate the first divergence. Circle the exact line (or lines) where the chatbot's reasoning first goes wrong. There may be more than one error.

A4. Test a small case. Plug $n = 2$ into both the claim and the chatbot's final expression. Do they agree? Show your arithmetic.

A5. **Classify the failure mode.** Write it in the margin next to the circled line: axiom reversion, invalid syntactic shortcut, or spatial / counting miscount.

A6. **Explain why the failure was likely.** Discuss with your partner why the system prompt made this error probable. The system prompt forced the failure here, but the same failure mode can surface on a genuine problem with no such prompt; explain why in one or two sentences.

Synthesize A1-A6 into a one-paragraph diagnosis below.

Practice transcript 2

Problem given to the chatbot. How many binary strings of length 6 contain exactly two 1s, with no two 1s adjacent? —————

Chatbot response.

Choose the two positions for the 1s: $C(6,2) = 15$ ways. Now exclude the placements where the two 1s are adjacent. The adjacent position pairs are (1,2), (2,3), (3,4), (4,5), (5,6), and (6,7), so there are 6 adjacent placements. By the subtraction rule, the count is $15 - 6 = 9$ strings.

(6,7) & A

$\begin{array}{cccccc} _ & _ & _ & _ & _ & _ \\ 1 & 2 & 3 & 4 & 5 & 6 \end{array}$

Your diagnosis of the chatbot response using the autopsy procedure A1-A6.

A1: probably wrong

A2: $\underline{\binom{6}{2} = 15}$

$$\frac{6!}{2!4!} = \frac{6 \cdot 5}{2} = 15$$

$$\bar{A} = U - A$$

$$|\bar{A}| = |U| - |A|$$

$$|U| = \binom{6}{2}$$

$$\underline{|A| = 6}$$

$$\begin{aligned} |\bar{A}| &= 15 - 6 \\ &= 9 \end{aligned}$$

First divergence $(6, 7)$ & A
remove $(6, 7)$

$$|A - (6, 7)| = 5$$

$$|\bar{A}| = 15 - 5 = \underline{10}$$

Explain why you included $(6, 7)$, that doesn't
seem like a valid assignment

↑

AI Literacy Exercise B [Prompt Engineering Lab] (Group): Take three weak prompts, and rewrite each one using all four parts of a strong prompt. Run both the weak and the strong version in the course chatbot. Record what changed.

The four parts of a strong prompt

- B1. **Establish context.** Tell the chatbot which subject, question type, and goal.
- B2. **Require step-by-step justification.** Require each step to name the rule or prior line it uses.
- B3. **Build in self-questioning.** Before the chatbot finishes, have it list its assumptions and check them.
- B4. **Plan the corrective dialogue.** Know in advance what you will say if you spot an error.

Problem 1: Induction. Target problem: prove by induction that $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$ for all $n \geq 1$.

The weak prompt:

solve this: $1*2 + 2*3 + \dots + n(n+1) = ?$

Your stronger version, including planned follow up prompts based on potential output from chatbot. Label each strengthening technique you use as an inline citation:

Strong: Repeated target problem, show all work, explain steps used

Weak: didn't do it by induction, used guess form as answer

Strong: Went through the proof as if in class explained each step

Problem 2: Combinatorics. Target problem: how many 5-card hands from a standard 52-card deck contain exactly two pairs (two cards of one rank, two cards of a different rank, and one card of a third rank)?

The weak prompt:

how many two pair hands are there in poker

Your stronger version, including planned follow up prompts based on potential output from chatbot. Label each strengthening technique you use as an inline citation:

In a standard deck of cards, assuming 13 face cards, and 4 suits, how many 5 card hands containing 2 pairs (same face) are possible?

Both answers are the same

1st: gave answer in bold

strong: gave steps reevaluated what you had said

Problem 3: Predicate Calculus. Target problem: is the formula $\forall x \exists y P(x, y) \rightarrow \exists y \forall x P(x, y)$ valid? If yes, prove it. If no, give a counterexample interpretation.

The weak prompt:

is for all x there exists y equivalent to there exists y for all x

Your stronger version, including planned follow up prompts based on potential output from chatbot. Label each strengthening technique you use as an inline citation:

Weak: answer in bold
 2nd -pro: both gave same answer, same logic,
 pro gave rule to follow

Strong: Target prompt college level discrete math,
 goal is to determine validity, show all steps & explanation.
 List all assumptions & change them

After you run both versions. Fill in the following table for at least one of the three problems. Compare the weak and strong versions on four dimensions.

strong: produced chart summary
 →

Dimension	Weak prompt gave you...	Strong prompt gave you...
Correctness of final answer		
Quality of justification		
Something useful you learned		
Errors you had to correct		

AI Literacy Exercise C [Scaffolded Proof Collaboration] (Group): Solve a combinatorics problem you probably have not seen before, using the course chatbot as a thinking partner. **Every chatbot message must be answered with either a justification request or a correction** (labels C2 and C3 below, repeated as a loop). Do not accept a step without interrogating it.

The collaboration procedure (subgoal labels C1–C6)

- C1. **Frame the problem and set the rules.** Give context, state the problem exactly, forbid the final answer.
- C2. **Elicit one step with its rule named.** Ask for the next step and the counting rule it uses.
- C3. **Interrogate the claim.** Request a justification or issue a correction; never accept a step silently.
- C4. **Verify against hand-computed cases.** Small cases computed by hand decide who is right.
- C5. **Consolidate in your own words.** The write-up is yours, not the chatbot's.
- C6. **Question the system.** Ask who built it, on whose data, and why you trust its answer.

Labels C2 and C3 repeat as a loop, once per chatbot message. Labels C4 through C6 conclude the session.

An example of how to start. The example below uses a different problem: the number of ways to place 8 non-attacking rooks on a chessboard. Each student turn is tagged with the subgoal label it performs.

STUDENT [C1: frame the problem and set the rules]: I'm in CSC 226 working on combinatorics. I want to figure out how many ways 8 non-attacking rooks can be placed on a standard 8×8 chessboard. I have not seen this problem before. Please do NOT give me the final number. Instead, help me set up the counting argument step by step, and for each step tell me what rule of counting you're using and why.

CHATBOT: Two rooks attack each other if they share a row or a column. So for a valid placement, each row must contain exactly one rook, and each column must contain exactly one rook. Step 1 (product rule): choose which column the rook in row 1 goes in — 8 choices. Step 2: for row 2, the column used by row 1 is forbidden, so there are 7 choices. Continuing, row k has $(8 - k + 1)$ choices. The total is a product.

STUDENT [C3: interrogate the claim]: Justify the claim that each row must contain exactly one rook. Why couldn't a valid placement have a row with zero rooks and another row with two?

CHATBOT: A row with two rooks immediately has two rooks sharing that row, so they attack each other. For 8 rooks on 8 rows with no two sharing a row, we need at most one per row; with 8 rooks and 8 rows, each row must contain exactly one. The same argument works for columns.

STUDENT [C2: Elicit one step with its rule named]: This seems like we are counting permutations of these rooks. Explain to me whether this is equivalent to counting permutations of $\{1, \dots, 8\}$.

Your problem: binary strings without three consecutive 1s.

Let $a(n)$ denote the number of binary strings of length n that do not contain the substring "111". (That is: no three consecutive 1s anywhere in the string.) Find and prove a recurrence for $a(n)$. Then use it to compute $a(10)$.

Write your first prompt (label C1). Provide context (CSC 226, combinatorics), state the problem exactly, explicitly forbid the chatbot from giving you the answer, ask for the first step of a counting argument with rule-naming, and have a plan for how you will verify small cases by hand.

- + Do not provide the worked solution all at once. Only provide the solution one step at a time.
- + I am trying to learn this for my final exam in CS226 at NC State
- + provide a rationale on recurrence relations first
+ show calculation of $a(1)$ & $a(2)$

During the conversation. Run labels C2 and C3 on every chatbot message: elicit one step with its rule named, then interrogate it with a justification request or a correction. Move to the ledger below only when the chatbot has proposed a recurrence you have interrogated.

Verification ledger (label C4). Before accepting any recurrence the chatbot proposes, compute $a(n)$ for small n by hand and check that the recurrence reproduces your values. Fill in this table. If the chatbot's recurrence disagrees with your hand count at any row, stop and correct it. If every row agrees, verify one step of the chatbot's derivation anyway; agreement can conceal flawed reasoning.

n	$a(n)$ by hand	Strings you counted (or how you counted)	Chatbot's formula gives	Agree?
1				
2				
3				
4				
5				
6				

Your final write-up (label C5). In your own words (not the chatbot's), write the recurrence, a proof of its correctness, and the value of $a(10)$.

Step back: question the system (label C6). You just used the chatbot as a thinking partner. Now turn the same scrutiny on the tool itself. In a few sentences, answer two of the following:

1. Who built this system, and what biases may arise from how it was designed?
2. Whose mathematics (whose notation, problems, and worked examples) is it likely most fluent in, and whose might it handle poorly?
3. Did its confident tone ever earn more trust than it had?
4. What would you want to know about a system like this before relying on it in a course that mattered?